Machine Learning — Unit 2 Exam Answers

Unit 2: Unsupervised Learning: Clustering (Syllabus Unit IV)

SPPU B.E. (AI & DS) 2019 Pattern, Course 417521 · Exam: 70 marks, 2.5 hours · Paper unit = question pair (Q1/Q2 → Unit 1, Q3/Q4 → Unit 2, Q5/Q6 → Unit 3, Q7/Q8 → Unit 4).

Each answer is written for **(X + 2) marks**, where X = highest marks ever asked. Sources cite your PDFs by file name and PDF page number.

1. Unit Snapshot

No.	Question	Max Marks	Years Asked	Importance
U2-1	Explain the K-Means (centroid-based) clustering algorithm with example. State its advantages and disadvantages.	9	Nov-Dec 2023, Nov-Dec 2024, Nov-Dec 2025	★★★
U2-2	Explain the DBSCAN algorithm. How does it work? State its advantages and disadvantages.	9	Nov-Dec 2024, Nov-Dec 2025	★★★
U2-3	Describe the Gaussian Mixture Model (GMM) for distribution-based clustering with example. How is it different from other clustering algorithms?	9	Nov-Dec 2023, Nov-Dec 2025	★★★
U2-4	What is clustering? Elaborate the types of clustering.	9	Nov-Dec 2024	★
U2-5	Elaborate the need of clustering and explain how the elbow method is used to decide the value of k.	9	Nov-Dec 2023	★
U2-6	Differentiate Agglomerative Hierarchical Clustering and Divisive Hierarchical Clustering.	8	Nov-Dec 2024	★
U2-7	Explain the Divisive Hierarchical Clustering (DHC) algorithm with example.	8	Nov-Dec 2023	★
U2-8	Discuss applications of clustering in market segmentation, social network analysis, image segmentation and anomaly detection, with examples.	8	Nov-Dec 2025	★

2. Must-Do First (★★★)

1. **U2-1** — Explain the K-Means (centroid-based) clustering algorithm with example. State its advantages and disadvantages. (9 M)
2. **U2-2** — Explain the DBSCAN algorithm. How does it work? State its advantages and disadvantages. (9 M)
3. **U2-3** — Describe the Gaussian Mixture Model (GMM) for distribution-based clustering with example. How is it different from other clustering algorithms? (9 M)

3. Full Answers

U2-1. Explain the K-Means (centroid-based) clustering algorithm with example. State its advantages and disadvantages.

Importance: ★★★ | **Max asked:** 9 M → **write for 11 M** | **Asked in:** Nov-Dec 2023, Nov-Dec 2024, Nov-Dec 2025

Source: ML Unit 3-6.pdf, pp. 45 and 52–58 (partitioning/centroid methods; K-Means and K-Medoids; book pp. 4-6, 4-13 to 4-19).

Asked as: *Explain K - Means clustering algorithm and states the advantages and disadvantages of k-means clustering algorithm.* (Nov-Dec 2023 Q3(a), 9 M) | *Describe centroid based clustering algorithm and explain any one type with example.* (Nov-Dec 2024 Q4(b), 9 M) | *Explain the K-Means clustering algorithm. What are its advantages and disadvantages.* (Nov-Dec 2025 Q4(b), 8 M)

Definition

K-Means is an **unsupervised, centroid-based (partitioning) clustering** algorithm that divides n data points into **K clusters**, where each point belongs to the cluster with the **nearest centroid (mean)**. It minimises the **Within-Cluster Sum of Squares (WCSS)**.

1. Centroid-Based Clustering

- Each cluster is represented by a **central vector (centroid)**; points are assigned to the closest centroid.
- K must be given **in advance**; it is an **iterative**, **flat (non-hierarchical)** method.
- Types: **K-Means** (centroid = mean), **K-Medoids/PAM** (centroid = actual data point), **K-Means++** (smart initialisation), Fuzzy C-Means.

2. K-Means Algorithm

1. Choose the number of clusters **K**.
2. **Initialise** K centroids randomly (or with K-Means++).
3. **Assign** each point to the **nearest centroid** (Euclidean distance).
4. **Update** each centroid = **mean** of points assigned to it.
5. **Repeat** steps 3–4 until centroids **do not change** (convergence) or max iterations reached.

$$\text{Objective: } J = \Sigma_k \Sigma_{\{x \in C_k\}} \|x - \mu_k\|^2$$

- C_k = k-th cluster, μ_k = its centroid, $\|x - \mu_k\|$ = Euclidean distance.

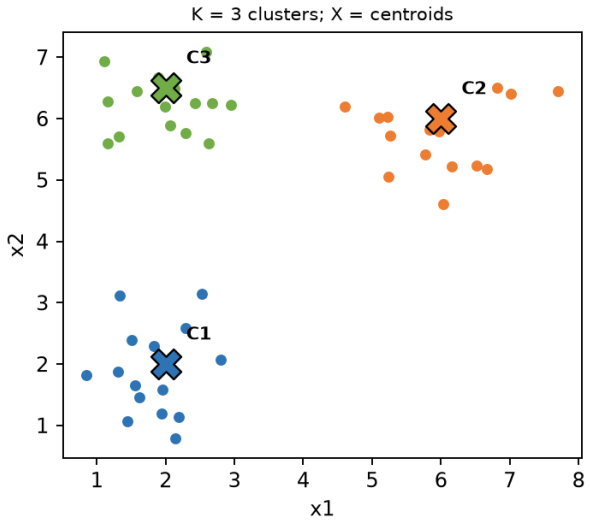


Fig. K-Means with K = 3: points grouped around centroids C1, C2, C3

- **Coloured points:** members of each cluster. **X marks:** centroids (means). Boundaries lie midway between centroids.

3. Worked Example (1-D, K = 2)

Data: {2, 4, 10, 12, 3, 20, 30, 11, 25}; initial centroids $\mathbf{m_1 = 2, m_2 = 4}$.

Iteration	Cluster 1	Cluster 2	New m_1	New m_2
1	{2, 3}	{4, 10, 12, 20, 30, 11, 25}	2.5	16
2	{2, 3, 4}	{10, 12, 20, 30, 11, 25}	3	18
3	{2, 3, 4, 10}	{12, 20, 30, 11, 25}	4.75	19.6
4	{2, 3, 4, 10, 12, 11}	{20, 30, 25}	7	25
5	{2, 3, 4, 10, 12, 11}	{20, 30, 25}	7	25 (no change → stop)

Final clusters: {2, 3, 4, 10, 11, 12} and {20, 25, 30}. (In iteration 1, point 3 is equidistant from 2 and 4; the tie is given to cluster 1.)

4. Advantages

- **Simple**, easy to implement and interpret.
- **Fast & scalable** – O(n·K·t); works on large datasets.
- Guaranteed to **converge**; tight, spherical clusters.

5. Disadvantages

- **K must be specified** in advance (use elbow/silhouette).
- **Sensitive to initial centroids** (different runs → different results) and to **outliers**.
- Assumes **spherical, equal-size** clusters; fails on non-convex shapes.
- Works only with **numerical** data (mean must exist).

6. Applications

- Customer segmentation, image compression (colour quantisation), document clustering.

Conclusion

K-Means repeatedly **assigns** points to the nearest mean and **updates** the means until stable; it is fast and popular but needs K and good initialisation.

Memory trick – "CAUR": Choose K → **A**ssign → **U**pdate → **R**epeat. Common mistake: forgetting the stopping condition.

U2-2. Explain the DBSCAN algorithm. How does it work? State its advantages and disadvantages.

Importance: ★★★ | **Max asked:** 9 M → **write for 11 M** | **Asked in:** Nov-Dec 2024, Nov-Dec 2025

Source: ML Unit 3-6.pdf, pp. 61–62 (density-based clustering, DBSCAN; book pp. 4-22, 4-23).

Asked as: *Explain DBSCAN algorithm with advantages and disadvantages.* (Nov-Dec 2024 Q4(a), 8 M) | *Explain the DBSCAN algorithm. How does it work? What are its advantages and disadvantages.* (Nov-Dec 2025 Q3(a), 9 M)

Definition

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) groups points that lie in **dense regions** and marks points in **low-density regions as noise (outliers)**. It needs two parameters: **ε (eps)** – neighbourhood radius and **MinPts** – minimum points to form a dense region.

1. Key Terms

- **ε-neighbourhood:** all points within distance ε of a point.
- **Core point:** has **≥ MinPts** points (including itself) in its ε-neighbourhood.
- **Border point:** fewer than MinPts neighbours but lies in the ε-neighbourhood of a core point.
- **Noise point:** neither core nor border.
- **Directly density-reachable:** q is in ε-neighbourhood of core point p.
- **Density-reachable:** reachable through a chain of core points. **Density-connected:** both reachable from a common core point.

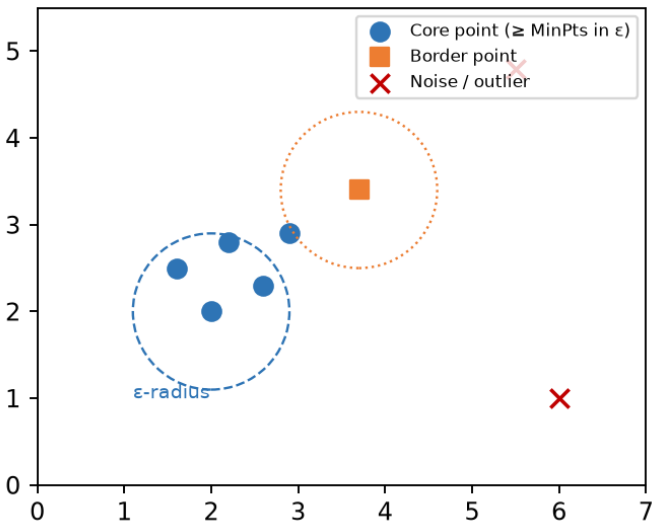


Fig. DBSCAN: core, border and noise points with ε-radius

- **Blue (dashed circle):** core point and its ε-radius containing ≥ MinPts points.
- **Orange square:** border point (inside a core's ε but not dense itself).
- **Red ×:** noise / outliers – not assigned to any cluster.

2. Algorithm / Working

1. Set **ε** and **MinPts**; mark all points unvisited.
2. Pick an **unvisited point p**; mark it visited and find its ε-neighbours.
3. If neighbours **< MinPts** → mark p as **noise** (may later become border).
4. If neighbours **≥ MinPts** → p is a **core point**; start a **new cluster** and add all neighbours.
5. **Expand:** for each neighbour that is also a core point, add its neighbours to the cluster (density-reachable points).
6. Repeat until all points are visited. Clusters = **density-connected** sets.

3. Example

Customer GPS locations: dense shopping areas form clusters of any shape (along roads); isolated locations are flagged as **noise**. With ε = 1 km, MinPts = 5.

4. Advantages

- **No need to specify the number of clusters.**
- Finds clusters of **arbitrary shape** (non-spherical).
- **Robust to outliers** – identifies noise explicitly.
- Only two parameters; order of data has little effect.

5. Disadvantages

- Choosing **ε and MinPts** is difficult (k-distance graph helps).
- Fails when clusters have **varying densities**.
- Poor in **high dimensions** (distance becomes less meaningful); $O(n^2)$ without index.
- Border points may be assigned to different clusters depending on order.

Conclusion

DBSCAN clusters by **density**, finding arbitrary-shaped clusters and outliers automatically, but is sensitive to ε and MinPts.

Memory trick – "C-B-N": **C**ore (dense), **B**order (edge), **N**oise (alone). "Two knobs: **ε** = how far, **MinPts** = how many."

U2-3. Describe the Gaussian Mixture Model (GMM) for distribution-based clustering with example. How is it different from other clustering algorithms?

Importance: ★★★ | **Max asked:** 9 M → **write for 11 M** | **Asked in:** Nov-Dec 2023, Nov-Dec 2025

Source: ML Unit 3-6.pdf, pp. 64–65 (Sec. 4.6 distribution-based clustering, 4.6.1 Gaussian Mixture Model, EM; book pp. 4-25, 4-26).

Asked as: *Explain Gaussian mixture model with example.* (Nov-Dec 2023 Q3(b), 8 M) | *Describe the Gaussian Mixture Model (GMM) for distribution-based clustering. How is it different from other clustering algorithms?* (Nov-Dec 2025 Q4(a), 9 M)

Definition

A **Gaussian Mixture Model (GMM)** is a **probabilistic, distribution-based clustering** model which assumes data is generated from a **mixture of K Gaussian (normal) distributions**, each with its own **mean μ_k, covariance Σ_k and mixing weight π_k**. It performs **soft clustering** – each point gets a **probability** of belonging to each cluster.

1. Model

$$p(x) = \Sigma_k \, \pi_k \cdot N(x \mid \mu_k, \Sigma_k) \, , \;\; \Sigma_k \, \pi_k = 1$$

- π_k = mixing coefficient (prior weight of cluster k), μ_k = mean (centre), Σ_k = covariance (shape/orientation), **N** = Gaussian density.

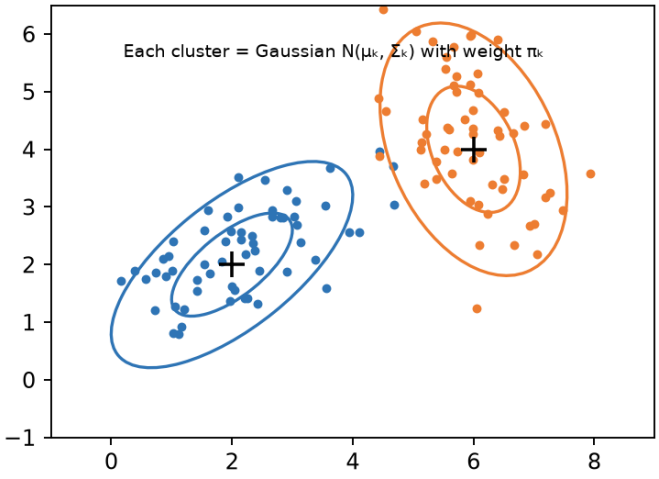


Fig. GMM: two Gaussian components with elliptical contours

- **Ellipses:** 1σ and 2σ contours of each Gaussian – clusters can be **elliptical** and of different sizes.
- **+ marks:** means μ_k. Points in the overlap get split membership probabilities.

2. EM Algorithm for GMM

1. **Initialise** μ_k, Σ_k, π_k (randomly or from K-Means).
2. **E-step (Expectation):** compute **responsibility** $\gamma(z_{nk})$ = probability that point x_n belongs to cluster k:

$$\gamma_{nk} = \pi_k \, N(x_n \mid \mu_k, \Sigma_k) \, / \, \Sigma_j \, \pi_j \, N(x_n \mid \mu_j, \Sigma_j)$$

3. **M-step (Maximisation):** update parameters using responsibilities:
   - $\mu_k = \Sigma_n \, \gamma_{nk} \, x_n \, / \, N_k$; $\Sigma_k = \Sigma_n \, \gamma_{nk} \, (x_n - \mu_k)(x_n - \mu_k)^T \, / \, N_k$; $\pi_k = N_k \, / \, N$, where $N_k = \Sigma_n \, \gamma_{nk}$.
4. Compute **log-likelihood**; repeat E and M steps until it **converges**.
5. Assign each point to the cluster with the **highest probability** (if hard labels needed).

3. Example

Heights of a mixed group of adults: two overlapping bell curves – women ($\mu \approx 160$ cm) and men ($\mu \approx 175$ cm). GMM learns both Gaussians; a person of 168 cm may get **0.45 women / 0.55 men**.

4. How GMM differs from other clustering algorithms

Basis	GMM	K-Means / DBSCAN / Hierarchical
Assignment	**Soft** (probabilities)	**Hard** (one cluster)
Cluster shape	**Elliptical**, different sizes (covariance)	K-Means: spherical; DBSCAN: arbitrary
Basis	**Statistical distribution** (likelihood)	Distance (K-Means, HC) or density (DBSCAN)
Algorithm	**EM** (E-step, M-step)	Assign–update / region growing / merge-split
Output	μ, Σ, π + membership probabilities	Labels (and centroids)
Generative	Yes – can generate new samples	No
K required	Yes (choose by BIC/AIC)	K-Means yes; DBSCAN no

5. Advantages / Disadvantages

- **+** Flexible cluster shapes; soft membership; probabilistic interpretation; density estimation.
- **–** Needs K; sensitive to initialisation; can converge to local optimum; slower than K-Means; assumes Gaussian data.

Conclusion

GMM models data as a **weighted sum of Gaussians** learnt by **EM**, giving **soft, elliptical** clusters – a probabilistic generalisation of K-Means.

Memory trick: "**E** = Estimate who belongs, **M** = Move the Gaussians." GMM = "**soft K-Means with shapes**".

U2-4. What is clustering? Elaborate the types of clustering.

Importance: ★ | **Max asked:** 9 M → **write for 11 M** | **Asked in:** Nov-Dec 2024

Source: ML Unit 3-6.pdf, pp. 41–48, 61, 64 (Sec. 4.1 What is clustering; types: partitioning, hierarchical, density, distribution; book pp. 4-2 to 4-9, 4-22, 4-25). Grid-based and fuzzy types are standard additions.

Asked as: *What is clustering? Elaborate Types of Clustering.* (Nov-Dec 2024 Q3(b), 9 M)

Definition

Clustering is an **unsupervised learning** technique that groups **unlabelled** data objects so that objects in the **same cluster are similar** (high intra-cluster similarity) and objects in **different clusters are dissimilar** (low inter-cluster similarity).

1. Characteristics of good clustering

- High **intra-cluster** similarity, low **inter-cluster** similarity.
- Scalable, handles different attribute types, noise and arbitrary shapes; interpretable.

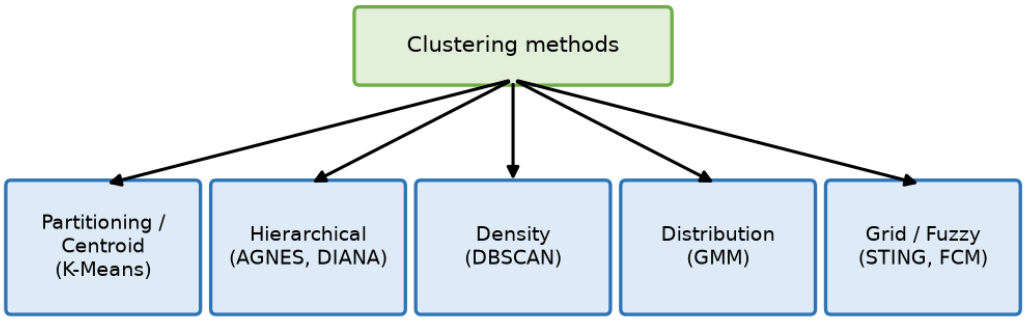


Fig. Types of clustering methods

2. Types of Clustering

1. **Partitioning / Centroid-based clustering**
   - Divides data into **K non-overlapping** groups; each represented by a centroid.
   - Examples: **K-Means, K-Medoids (PAM), CLARA**.
   - Fast; needs K; spherical clusters.
2. **Hierarchical (connectivity-based) clustering**
   - Builds a **tree of clusters (dendrogram)**.
   - **Agglomerative (AGNES)** – bottom-up merging; **Divisive (DIANA)** – top-down splitting.
   - No need for K in advance; costly $O(n^2)$.
3. **Density-based clustering**
   - Clusters = **dense regions** separated by sparse regions; finds arbitrary shapes and noise.
   - Examples: **DBSCAN, OPTICS, Mean-shift**.
4. **Distribution (model)-based clustering**
   - Assumes data comes from **probability distributions**; soft membership.
   - Example: **Gaussian Mixture Model (EM)**.
5. **Grid-based clustering**
   - Quantises space into a **grid of cells** and clusters the cells; very fast.
   - Examples: **STING, CLIQUE**.
6. **Fuzzy (soft) clustering**
   - A point can belong to **several clusters with membership degrees** (0–1).
   - Example: **Fuzzy C-Means**.

- *Also:* **Hard vs soft** clustering; **flat vs hierarchical**.

3. Example

Online store groups customers by spending and visits: K-Means → 3 segments (budget, regular, premium); DBSCAN flags bulk-buying outliers.

Type	Idea	Example
Partitioning	Nearest centroid	K-Means
Hierarchical	Merge / split tree	AGNES, DIANA
Density	Dense regions	DBSCAN
Distribution	Mixture of Gaussians	GMM
Grid	Cells of space	STING
Fuzzy	Membership degrees	FCM

Conclusion

Clustering finds natural groups in unlabelled data; the type is chosen according to cluster **shape, size, noise and data volume**.

Memory trick – "PHD-DG-F": **P**artitioning, **H**ierarchical, **D**ensity, **D**istribution, **G**rid, **F**uzzy.

U2-5. Elaborate the need of clustering and explain how the elbow method is used to decide the value of k.

Importance: ★ | **Max asked:** 9 M → **write for 11 M** | **Asked in:** Nov-Dec 2023

Source: ML Unit 3-6.pdf, pp. 41–42 (uses / need of cluster analysis) and pp. 59–60 (elbow method; book pp. 4-2, 4-3, 4-20, 4-21).

Asked as: *Elaborate need of clustering and explain how the elbow method is used to decide the value of cluster k.* (Nov-Dec 2023 Q4(a), 9 M)

Definition

Clustering groups unlabelled data into meaningful groups. The **Elbow method** is a graphical technique to choose the **optimal number of clusters K** for K-Means by plotting **WCSS** against K and picking the "**elbow**" point.

1. Need of Clustering

1. **Labelled data is costly** – clustering finds structure without labels.
2. **Pattern discovery:** reveals hidden groups (customer types, gene families).
3. **Data summarisation / reduction:** represent many points by few centroids (compression).
4. **Pre-processing** for other algorithms (features, sampling, semi-supervised labelling).
5. **Outlier/anomaly detection:** points far from all clusters.
6. **Decision making:** targeted marketing, resource allocation, recommendation.

2. Elbow Method – Steps

1. Run K-Means for K = 1, 2, …, 10.
2. For each K compute **WCSS** (inertia):

$$\text{WCSS} = \Sigma_k \, \Sigma_\{x \in C_k\} \, \|x - \mu_k\|^2$$

   - μ_k = centroid of cluster C_k; $\|\cdot\|$ = Euclidean distance.
3. **Plot** WCSS (y-axis) vs K (x-axis).
4. WCSS always decreases as K increases (K = n → WCSS = 0).
5. Choose K at the **"elbow"** – where the decrease **sharply slows down**; adding more clusters gives little gain.

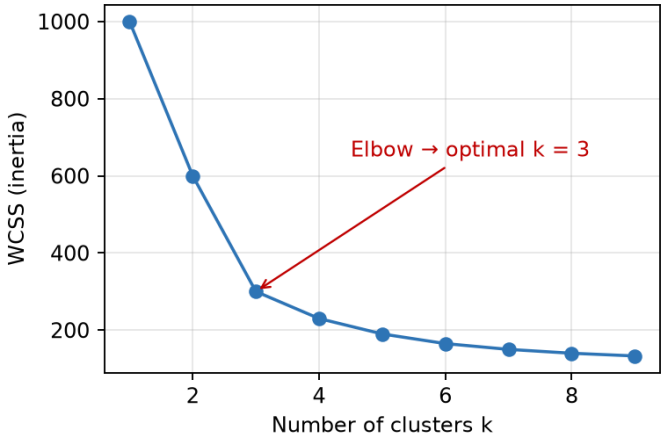


Fig. Elbow curve: WCSS drops sharply till K = 3, then flattens

- **x-axis:** number of clusters K; **y-axis:** WCSS.
- **Elbow (K = 3):** best trade-off between compactness and number of clusters.

3. Example

WCSS = 1000, 600, 300, 230, 190 for K = 1…5 → drop 400, 300, then only 70 → **K = 3**.

4. Limitations

- Elbow may be **unclear/smooth**; then use **silhouette score** or gap statistic.

Conclusion

Clustering is needed to discover structure in unlabelled data, and the elbow method gives a simple visual way to choose K for K-Means.

Memory trick: "Bend your **arm** – the **elbow** is where the curve bends." Plot **WCSS vs K**, not accuracy.

U2-6. Differentiate Agglomerative Hierarchical Clustering and Divisive Hierarchical Clustering.

Importance: ★ | **Max asked:** 8 M → **write for 10 M** | **Asked in:** Nov-Dec 2024

Source: ML Unit 3-6.pdf, pp. 48–52 (hierarchical clustering: agglomerative and divisive, dendrogram; book pp. 4-9 to 4-13). The worked example is a standard illustration.

Definition

Hierarchical clustering builds a **hierarchy (tree) of clusters** shown as a **dendrogram**. **Agglomerative (AHC/AGNES)** works **bottom-up** by merging; **Divisive (DHC/DIANA)** works **top-down** by splitting.

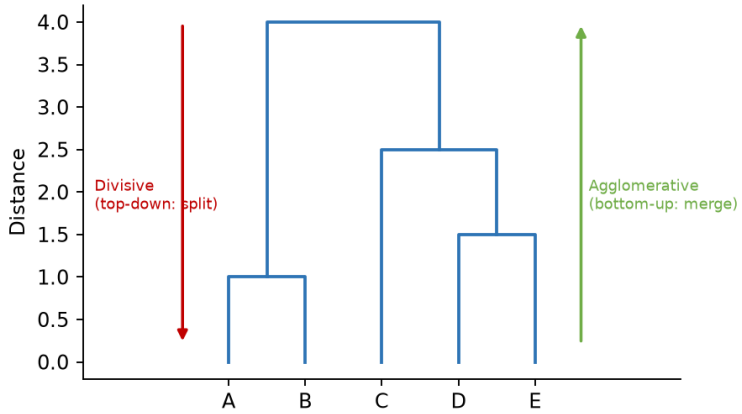


Fig. Dendrogram: agglomerative merges upward, divisive splits downward

Basis	Agglomerative (AHC)	Divisive (DHC)
Approach	**Bottom-up**	**Top-down**
Start	Each point is its **own cluster** (n clusters)	**All points in one** cluster
Operation	**Merge** the two closest clusters	**Split** a cluster into two
End	One cluster (or K clusters)	Each point alone (or K clusters)
Decision based on	**Linkage**: single, complete, average, Ward	Largest **diameter** / dissimilarity (DIANA), or K-Means split (bisecting)
Complexity	$O(n^2 \log n)$ to $O(n^3)$	$O(2^n)$ exhaustive; heuristics needed
Accuracy	Local decisions; early merges cannot be undone	Uses **global view** first → can be more accurate at top levels
Popularity	**More common**, easy to implement	Less common, complex
Algorithm name	AGNES	DIANA
Good for	Finding small clusters	Finding large clusters

Example

Points A–E: AHC merges (A,B), (D,E), then C with (D,E), then all. DHC first splits {A..E} into {A,B} and {C,D,E}, then continues splitting.

Conclusion

Both produce a dendrogram; **AHC merges** small clusters upward, **DHC splits** the whole set downward. Cut the dendrogram at a height to get K clusters.

Memory trick: Agglomerative = **A**dd together (bottom-up); **D**ivisive = **D**ivide (top-down).

U2-7. Explain the Divisive Hierarchical Clustering (DHC) algorithm with example.

Importance: ★ | **Max asked:** 8 M → **write for 10 M** | **Asked in:** Nov-Dec 2023

Source: ML Unit 3-6.pdf, pp. 48–52 (hierarchical clustering: agglomerative and divisive, dendrogram; book pp. 4-9 to 4-13). The worked example is a standard illustration.

Asked as: *Explain Divisive Hierarchical clustering (DHC) algorithm with example.* (Nov-Dec 2023 Q4(b), 8 M)

Definition

Divisive Hierarchical Clustering (DHC / DIANA – DIvisive ANAlysis) is a **top-down** clustering method: it starts with **all points in one cluster** and **recursively splits** clusters until each point is a separate cluster or a stopping criterion (K clusters) is met.

1. Algorithm (DIANA)

1. Start with **one cluster** containing all n points; compute the **distance matrix**.
2. Choose the cluster with the **largest diameter** (max distance between any two of its points).
3. In that cluster, find the point with the **highest average dissimilarity** to the others → it starts a new **"splinter group"**.
4. **Move** points that are closer (on average) to the splinter group than to the remaining group into the splinter group.
5. Repeat step 4 until no point moves → the cluster is split into **two**.
6. Repeat steps 2–5 until every cluster has **one point** (or K clusters reached).
7. Draw the **dendrogram** (top-down).

- *Alternative:* **Bisecting K-Means** – split the largest cluster with K-Means (K = 2) repeatedly.

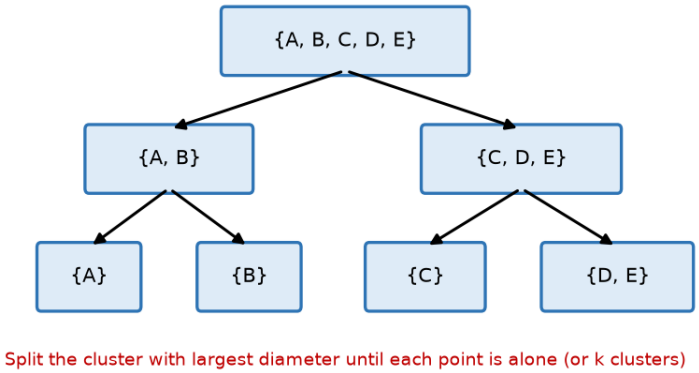


Fig. Divisive clustering: splitting {A..E} step by step

2. Example

1-D points: A = 1, B = 2, C = 6, D = 9, E = 10.

- **Step 1:** {A,B,C,D,E}. Average distance of A to others = (1+5+8+9)/4 = **5.75** (highest) → A starts splinter group.
- B: avg to splinter {A} = 1, to rest {C,D,E} = (4+7+8)/3 = 6.33 → **B moves**. C: to {A,B} = 4.5, to {D,E} = 3.5 → stays.
- **Split 1:** {A, B} and {C, D, E}.
- **Step 2:** largest diameter is {C,D,E} (10 − 6 = 4) → C is farthest (avg 3.5) → **{C}** and **{D, E}**.
- **Step 3:** split {A,B} → {A},{B}; split {D,E} → {D},{E}. Done.

3. Advantages / Disadvantages

- **+** Takes a **global view** first; good at finding **large clusters**; no K needed (cut dendrogram).
- **−** Computationally **expensive**; splits cannot be undone; sensitive to outliers.

Conclusion

DHC (DIANA) starts from one cluster and keeps **splitting the widest cluster** using the splinter-group idea until all points are separated, producing a top-down dendrogram.

Memory trick: "**DIANA divides** – find the **odd one out** (splinter), pull its friends, split."

U2-8. Discuss applications of clustering in market segmentation, social network analysis, image segmentation and anomaly detection, with examples.

Importance: ★ | **Max asked:** 8 M → **write for 10 M** | **Asked in:** Nov-Dec 2025

Source: ML Unit 3-6.pdf, pp. 66–67 (Sec. 4.7 Application of Clustering Technique: market segmentation, social network analysis, image segmentation, anomaly detection; book pp. 4-27, 4-28).

Asked as: *Discuss the applications of clustering techniques in market segmentaion, social network analysis, image segmentation, and anomaly detection. Provide examples for each application.* (Nov-Dec 2025 Q3(b), 8 M)

Definition

Clustering groups similar unlabelled objects; because it needs **no labels**, it is widely used to discover segments, communities, regions and outliers in real data.

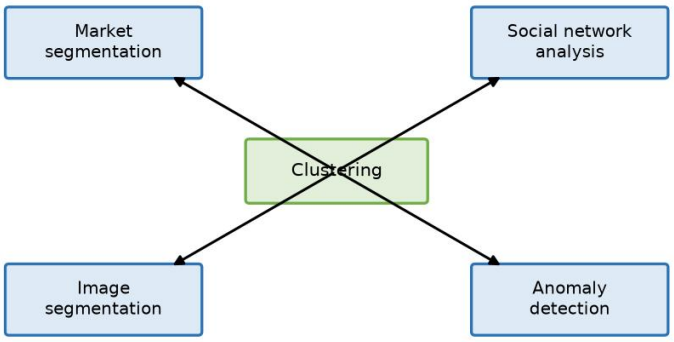


Fig. Major applications of clustering

1. Market Segmentation

- Group customers by **demographics, purchase history, spending, frequency** (RFM).
- Algorithm: **K-Means / GMM**.
- **Example:** a supermarket finds 3 segments – *budget shoppers, regular families, premium buyers* – and sends targeted offers to each.

2. Social Network Analysis

- Detect **communities** – groups of users who interact more with each other than with others.
- Algorithm: **hierarchical / graph (spectral) clustering**.
- **Example:** Facebook/LinkedIn suggest friends or groups by clustering users with common connections and interests; detecting fake-account rings.

3. Image Segmentation

- Group **pixels** with similar colour, intensity or texture into regions.
- Algorithm: **K-Means on pixel colours**, mean-shift, DBSCAN.
- **Example:** in an MRI scan, separating **tumour tissue** from healthy tissue; separating road, sky and vehicles in self-driving car images; colour quantisation.

4. Anomaly (Outlier) Detection

- Points that **do not belong to any cluster** or lie **far from centroids** are anomalies.

- Algorithm: **DBSCAN (noise points)**, distance from K-Means centroid.
- **Example:** credit-card **fraud** – a transaction far from a customer's usual cluster (amount, location, time) is flagged; network intrusion detection.

5. Other Applications (brief)

- Document/news grouping, recommendation systems, biology (gene expression), city planning, insurance.

Application	What is clustered	Typical algorithm	Example
Market segmentation	Customers	K-Means, GMM	Budget vs premium buyers
Social network analysis	Users / nodes	Hierarchical, spectral	Friend communities
Image segmentation	Pixels	K-Means, mean-shift	Tumour region in MRI
Anomaly detection	Transactions / events	DBSCAN	Card fraud

Conclusion

Clustering turns raw unlabelled data into **actionable groups** – segments for marketing, communities in networks, regions in images and outliers for security.

Memory trick – "MSIA": Market, **S**ocial, **I**mage, **A**nomaly. Always give one concrete example per application.

4. Last-Minute Revision Sheet

Key Definitions

- **Clustering:** unsupervised grouping; high intra-, low inter-cluster similarity.
- **K-Means:** centroid-based; minimise WCSS; Choose K → Assign → Update → Repeat.
- **Elbow method:** plot WCSS vs K; choose bend point.
- **DBSCAN:** density-based; ε & MinPts; core / border / noise; arbitrary shapes; no K.
- **GMM:** mixture of Gaussians; soft clustering; EM algorithm (E: responsibilities, M: update μ, Σ, π).
- **Hierarchical:** dendrogram; AHC (AGNES, bottom-up merge), DHC (DIANA, top-down split).

Formulas

Item	Formula
Euclidean distance	$\sqrt{\Sigma(x_i - y_i)^2}$
K-Means objective / WCSS	$\Sigma_k \Sigma \|\|x - \mu_k\|\|^2$
Centroid update	μ_k = mean of points in C_k
GMM density	$p(x) = \Sigma\, \pi_k\, N(x \mid \mu_k, \Sigma_k)$, $\Sigma\pi_k = 1$
GMM responsibility	$\gamma_{nk} = \pi_k N(x_n \mid \mu_k, \Sigma_k) / \Sigma_j\, \pi_j N(x_n \mid \mu_j, \Sigma_j)$
DBSCAN core point	$\|N_\varepsilon(p)\| \geq$ MinPts

Memory Tricks

- K-Means = **CAUR**; Clustering types = **PHD-DG-F**; DBSCAN = **C-B-N**.
- GMM = "soft K-Means with shapes"; EM = **E**stimate, **M**ove.
- **A**gglomerative = **A**dd; **D**ivisive = **D**ivide. Applications = **MSIA**.
- In K-Means numericals, show each iteration as a table and stop when centroids repeat.

5. Assumptions & Gaps

- **Unit mapping:** Q3/Q4 of each paper are taken as Unit 2 (= syllabus Unit IV, Clustering).
- **2024 Q4(b)** "centroid based clustering, explain any one type" is merged with K-Means (U2-1) because K-Means is the type normally explained.
- The K-Means and DIANA worked examples are standard illustrations created for practice (not copied from a paper).
- All Unit 2 topics were found in ML Unit 3-6.pdf (Unit IV); no ⚠ flags.